

Wake Vision Challenge (Model Centric Track) - Technical Report

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Table 1. Aging Evolution search boundaries.

Boundary name	Value
Validation Error	0.3
PMU	300 KiB
Model Size	500 KiB
MAC	1.000.000

1 Methodology

To construct the final model for the challenge, we deployed the open-source tool μ NAS [1], a robust Neural Architecture Search (NAS) library developed in Python. μ NAS is designed to search for and identify architectures optimized in terms of Peak Memory Usage (PMU), model size and Multiply and Accumulate (MAC) operations. The pipeline welcomes diverse search spaces for neural architectures such as Multi-Layer Perceptrons (MLPs), 1D [2] and 2D Convolutional Neural Networks (CNNs).

For the purposes of the challenge, we constrained the μ NAS search on the 2D convolutional space, exploring it through Aging Evolution [3] strategy. The neural search algorithm was already provided by the original implementation. The output of μ NAS are Keras models which are then converted in Tensorflow Lite format. The original implementation does not take into account Quantization Aware Training (QAT) for the generated models. To address this issue, in our implementation we modified μ NAS to obtain models ready for post-training quantization, especially of the input and output layer. We set μ NAS for quick training sessions of 10 epochs per model for the neural search. Finally, we trained the best-performing model for an additional 100 epochs.

Table 1 presents the selected bounds (indicative, not strict) for the μ NAS Aging Evolution neural search. Fig. 1 depicts the final model architecture found by μ NAS. We report a test accuracy of roughly 75% with 270,38 KiB of total weights (Flash size), 33,67 KiB of activations (RAM size) and a total number the number of MAC complexity (MACC) of 693.818. This confirms the deployability of the obtained model.

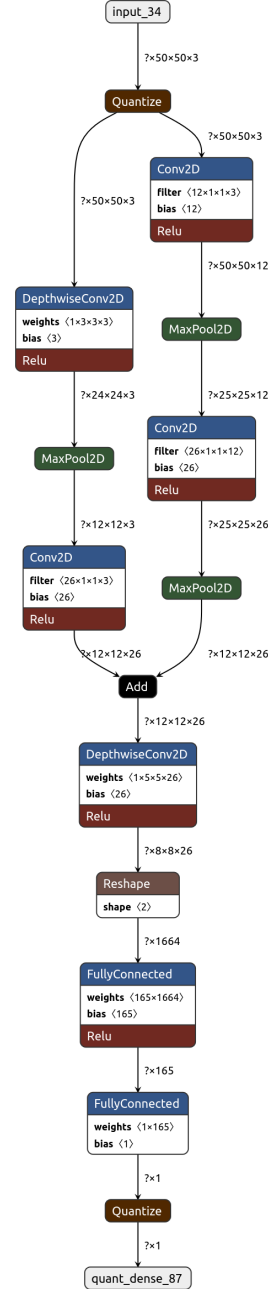


Figure 1. Final model architecture.

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References

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